

mielke-geometrodynamics: residuals

equations measured 2026-09-15T13:40:25Z against the model of 2026-09-15

Corrected (92)

Identifier	Page	Conf.	LaTeX source	Rendered	Scan image
mielke-geometrodynamics_EQ0006 (was)	33	■ 0.996	$\left.e_{\beta}\right\rceil\vartheta^{\alpha}=\delta^{\alpha}_{\beta}$.	$e_{\beta}\rfloor\vartheta^{\alpha}=\delta^{\alpha}_{\beta}$.	$e_{\beta}\rfloor\vartheta^{\alpha}=\delta^{\alpha}_{\beta}$.
mielke-geometrodynamics_EQ0006 (now)	33	—	$e_{\beta}\rfloor\vartheta^{\alpha}=\delta^{\alpha}_{\beta}$.	$e_{\beta}\rfloor\vartheta^{\alpha}=\delta^{\alpha}_{\beta}$.	$e_{\beta}\rfloor\vartheta^{\alpha}=\delta^{\alpha}_{\beta}$.
basis: census / census+ink					
mielke-geometrodynamics_EQ0119 (was)	59	■ 0.646	$\begin{aligned} \Sigma_{\alpha} & \& \left.\left.:=e_{\alpha}\right\rceil L-\left(e_{\alpha}\right\rceil d A\right) \wedge \frac{\partial L}{\partial(d A)} \\ & =\frac{1}{2}\left[\left(e_{\alpha}\right\rceil F\right) \wedge H-F \wedge\left(e_{\alpha}\right\rceil H\right] \end{aligned}$	$\Sigma_{\alpha}:=e_{\alpha}\rfloor L-\left(e_{\alpha}\rfloor d A\right) \wedge \frac{\partial L}{\partial(d A)} \\ =\frac{1}{2}\left[\left(e_{\alpha}\rfloor F\right) \wedge H-F \wedge\left(e_{\alpha}\rfloor H\right)\right]$	$\Sigma_{\alpha}:=e_{\alpha}\rfloor L-\left(e_{\alpha}\rfloor d A\right) \wedge \frac{\partial L}{\partial(d A)} \\ =\frac{1}{2}\left[\left(e_{\alpha}\rfloor F\right) \wedge H-F \wedge\left(e_{\alpha}\rfloor H\right)\right]$
mielke-geometrodynamics_EQ0119 (now)	59	—	$\begin{aligned} \Sigma_{\alpha} & \& :=e_{\alpha}\rfloor L-\left(e_{\alpha}\rfloor d A\right) \wedge \frac{\partial L}{\partial(d A)} \\ & =\frac{1}{2}\left[\left(e_{\alpha}\rfloor F\right) \wedge H-F \wedge\left(e_{\alpha}\rfloor H\right)\right] \end{aligned}$	$\Sigma_{\alpha}:=e_{\alpha}\rfloor L-\left(e_{\alpha}\rfloor d A\right) \wedge \frac{\partial L}{\partial(d A)} \\ =\frac{1}{2}\left[\left(e_{\alpha}\rfloor F\right) \wedge H-F \wedge\left(e_{\alpha}\rfloor H\right)\right]$	$\Sigma_{\alpha}:=e_{\alpha}\rfloor L-\left(e_{\alpha}\rfloor d A\right) \wedge \frac{\partial L}{\partial(d A)} \\ =\frac{1}{2}\left[\left(e_{\alpha}\rfloor F\right) \wedge H-F \wedge\left(e_{\alpha}\rfloor H\right)\right]$
basis: census / census+ink					
mielke-geometrodynamics_EQ0120 (was)	60	■ 0.651	$d \Sigma_{\alpha} \simeq\left(e_{\alpha}\right\downarrow F) \wedge j$	$d \Sigma_{\alpha} \simeq\left(e_{\alpha}\rfloor F\right) \wedge j$	$d \Sigma_{\alpha} \simeq\left(e_{\alpha}\rfloor F\right) \wedge j$
mielke-geometrodynamics_EQ0120 (now)	60	—	$d \Sigma_{\alpha} \simeq\left(e_{\alpha}\rfloor F\right) \wedge j$	$d \Sigma_{\alpha} \simeq\left(e_{\alpha}\rfloor F\right) \wedge j$	$d \Sigma_{\alpha} \simeq\left(e_{\alpha}\rfloor F\right) \wedge j$
basis: census / census					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_100 EQ0292 (was)		■ 0.165	$\begin{aligned} \mathfrak{f}_{\xi} L &= \left(\mathfrak{f}_{\xi} g_{\alpha\beta} \right) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \left(\mathfrak{f}_{\xi} \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ \left(\mathfrak{f}_{\xi} \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\mathfrak{f}_{\xi} D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[\left(\xi \rfloor \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \rfloor D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- \left(\xi \rfloor \vartheta^{\alpha} \right) D \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor T^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \lrcorner R_{\beta}^{\gamma} \right) \wedge I^{\beta}{}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ \left(\xi \rfloor D\Psi \right) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p \left(\xi \rfloor \Psi \right) \wedge D \frac{\delta L}{\delta \Psi} . \end{aligned}$	$\begin{aligned} \mathfrak{f}_{\xi} L &= \left(\mathfrak{f}_{\xi} g_{\alpha\beta} \right) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \left(\mathfrak{f}_{\xi} \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ \left(\mathfrak{f}_{\xi} \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\mathfrak{f}_{\xi} D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[\left(\xi \rfloor \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \rfloor D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- \left(\xi \rfloor \vartheta^{\alpha} \right) D \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor T^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \lrcorner R_{\beta}^{\gamma} \right) \wedge I^{\beta}{}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ \left(\xi \rfloor D\Psi \right) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p \left(\xi \rfloor \Psi \right) \wedge D \frac{\delta L}{\delta \Psi} . \end{aligned}$	$\begin{aligned} \mathfrak{f}_{\xi} L &= \left(\mathfrak{f}_{\xi} g_{\alpha\beta} \right) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \left(\mathfrak{f}_{\xi} \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ \left(\mathfrak{f}_{\xi} \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\mathfrak{f}_{\xi} D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[\left(\xi \rfloor \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \rfloor D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- \left(\xi \rfloor \vartheta^{\alpha} \right) D \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor T^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \lrcorner R_{\beta}^{\gamma} \right) \wedge I^{\beta}{}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ \left(\xi \rfloor D\Psi \right) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p \left(\xi \rfloor \Psi \right) \wedge D \frac{\delta L}{\delta \Psi} . \end{aligned}$
Conf. MathPix confidence: ■ ≥ 0.9 ■ $0.5-0.9$ ■ < 0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_E00292 (now)	100	—	$\begin{aligned} \mathfrak{f}_{\xi} L &\equiv \left(\mathfrak{f}_{\xi} g_{\alpha\beta} \right) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \left(\mathfrak{f}_{\xi} \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ \left(\mathfrak{f}_{\xi} \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\mathfrak{f}_{\xi} D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[\left(\xi \rfloor \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \rfloor D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- \left(\xi \rfloor \vartheta^{\alpha} \right) D \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor T^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor R_{\beta}^{\gamma} \right) \wedge I^{\beta}{}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ \left(\xi \rfloor D\Psi \right) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p \left(\xi \rfloor \Psi \right) \wedge D \frac{\delta L}{\delta \Psi} . \end{aligned}$	$\begin{aligned} \mathfrak{f}_{\xi} L &= \left(\mathfrak{f}_{\xi} g_{\alpha\beta} \right) \wedge \frac{\partial L}{\partial g_{\alpha\beta}} + \left(\mathfrak{f}_{\xi} \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} \\ &+ \left(\mathfrak{f}_{\xi} \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\mathfrak{f}_{\xi} D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \\ &= D \left[\left(\xi \rfloor \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \rfloor D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} \right] \\ &- \left(\xi \rfloor \vartheta^{\alpha} \right) D \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor T^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor R_{\beta}^{\gamma} \right) \wedge I^{\beta}{}_{\gamma} \Psi \wedge \frac{\partial L}{\partial D\Psi} \\ &+ \left(\xi \rfloor D\Psi \right) \wedge \frac{\delta L}{\delta \Psi} + (-1)^p \left(\xi \rfloor \Psi \right) \wedge D \frac{\delta L}{\delta \Psi} . \end{aligned}$	
basis: census / census					
mielke-geometrodynamics_E00293 (was)	101	■0.052	$\xi \downharpoonleft L \equiv \left(\xi \downharpoonleft \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} .$	$\xi \downharpoonleft L \equiv \left(\xi \downharpoonleft \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} .$	
mielke-geometrodynamics_E00293 (now)	101	—	$\xi \rfloor L \equiv \left(\xi \rfloor \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \rfloor D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} .$	$\xi \rfloor L \equiv \left(\xi \rfloor \vartheta^{\alpha} \right) \wedge \frac{\partial L}{\partial \vartheta^{\alpha}} + \left(\xi \rfloor \Psi \right) \wedge \frac{\partial L}{\partial \Psi} + \left(\xi \rfloor D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} .$	
basis: census / census					
mielke-geometrodynamics_E00294 (was)	101	■0.510	$\left. \Sigma_{\alpha} \right L - \left(e_{\alpha} \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} - \left(e_{\alpha} \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} .$	$\Sigma_{\alpha} = e_{\alpha} \downharpoonleft L - \left(e_{\alpha} \downharpoonleft D\Psi \right) \wedge \frac{\partial L}{\partial D\Psi} - \left(e_{\alpha} \downharpoonleft \Psi \right) \wedge \frac{\partial L}{\partial \Psi} .$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_101 E08294 (now)	101	—	$\Sigma_{\alpha} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi}.$	$\Sigma_{\alpha} = e_{\alpha} \rfloor L - (e_{\alpha} \rfloor D \Psi) \wedge \frac{\partial L}{\partial D \Psi} - (e_{\alpha} \rfloor \Psi) \wedge \frac{\partial L}{\partial \Psi}.$	
basis: census / census+ink					
mielke-geometrodynamics_101 E08295 (was)	101	■0.850	$\begin{aligned} D \Sigma_{\alpha} &\wedge \left(e_{\alpha} \rfloor T^{\beta} \right) \wedge \left(e_{\alpha} \rfloor R^{\beta \gamma} \right) \wedge \tau_{\beta \gamma} + F_{\alpha} \\ &\simeq (e_{\alpha} \rfloor T^{\beta}) \wedge \Sigma_{\beta} + (e_{\alpha} \rfloor R^{\beta \gamma}) \wedge \tau_{\beta \gamma}. \end{aligned}$	$\begin{aligned} D \Sigma_{\alpha} &= (e_{\alpha} \rfloor T^{\beta}) \wedge \Sigma_{\beta} + (e_{\alpha} \rfloor R^{\beta \gamma}) \wedge \tau_{\beta \gamma} + F_{\alpha} \\ &\simeq (e_{\alpha} \rfloor T^{\beta}) \wedge \Sigma_{\beta} + (e_{\alpha} \rfloor R^{\beta \gamma}) \wedge \tau_{\beta \gamma}. \end{aligned}$	
mielke-geometrodynamics_101 E08295 (now)	101	—	$\begin{aligned} D \Sigma_{\alpha} &\wedge \left(e_{\alpha} \rfloor T^{\beta} \right) \wedge \left(e_{\alpha} \rfloor R^{\beta \gamma} \right) \wedge \tau_{\beta \gamma} + F_{\alpha} \\ &\simeq (e_{\alpha} \rfloor T^{\beta}) \wedge \Sigma_{\beta} + (e_{\alpha} \rfloor R^{\beta \gamma}) \wedge \tau_{\beta \gamma}. \end{aligned}$	$\begin{aligned} D \Sigma_{\alpha} &= (e_{\alpha} \rfloor T^{\beta}) \wedge \Sigma_{\beta} + (e_{\alpha} \rfloor R^{\beta \gamma}) \wedge \tau_{\beta \gamma} + F_{\alpha} \\ &\simeq (e_{\alpha} \rfloor T^{\beta}) \wedge \Sigma_{\beta} + (e_{\alpha} \rfloor R^{\beta \gamma}) \wedge \tau_{\beta \gamma}. \end{aligned}$	
basis: census / census+ink					
mielke-geometrodynamics_101 E08296 (was)	101	■0.847	$F_{\alpha} = (e_{\alpha} \rfloor D \Psi) \frac{\delta L}{\delta \Psi} + (-1)^p (e_{\alpha} \rfloor \Psi) \wedge D \frac{\delta L}{\delta \Psi}$	$F_{\alpha} = (e_{\alpha} \rfloor D \Psi) \frac{\delta L}{\delta \Psi} + (-1)^p (e_{\alpha} \rfloor \Psi) \wedge D \frac{\delta L}{\delta \Psi}$	
mielke-geometrodynamics_101 E08296 (now)	101	—	$F_{\alpha} = (e_{\alpha} \rfloor D \Psi) \frac{\delta L}{\delta \Psi} + (-1)^p (e_{\alpha} \rfloor \Psi) \wedge D \frac{\delta L}{\delta \Psi}$	$F_{\alpha} = (e_{\alpha} \rfloor D \Psi) \frac{\delta L}{\delta \Psi} + (-1)^p (e_{\alpha} \rfloor \Psi) \wedge D \frac{\delta L}{\delta \Psi}$	
basis: census / census					
mielke-geometrodynamics_102 E08302 (was)	102	■0.950	$\begin{aligned} \widehat{D} \Sigma_{\alpha} &\wedge \left(e_{\alpha} \rfloor R_{\beta}^{\gamma} \right) \wedge \Delta^{\beta} \gamma \\ &- \frac{1}{2} (e_{\alpha} \rfloor Q_{\beta \gamma}) \wedge \sigma^{\beta \gamma} \end{aligned}$	$\widehat{D} \Sigma_{\alpha} \cong (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \Delta^{\beta} \gamma - \frac{1}{2} (e_{\alpha} \rfloor Q_{\beta \gamma}) \wedge \sigma^{\beta \gamma}$	
mielke-geometrodynamics_102 E08302 (now)	102	—	$\begin{aligned} \widehat{D} \Sigma_{\alpha} &\wedge \left(e_{\alpha} \rfloor R_{\beta}^{\gamma} \right) \wedge \Delta^{\beta} \gamma \\ &- \frac{1}{2} (e_{\alpha} \rfloor Q_{\beta \gamma}) \wedge \sigma^{\beta \gamma} \end{aligned}$	$\widehat{D} \Sigma_{\alpha} \cong (e_{\alpha} \rfloor R_{\beta}^{\gamma}) \wedge \Delta^{\beta} \gamma - \frac{1}{2} (e_{\alpha} \rfloor Q_{\beta \gamma}) \wedge \sigma^{\beta \gamma}$	
basis: census / census+ink					
mielke-geometrodynamics_103 E08305 (was)	103	■0.793	$\varepsilon_{\mathrm{RC}} := \xi^{\alpha} \Sigma_{\alpha} + \left(e_{\beta} \rfloor \widehat{D} \xi^{\gamma} \right) \tau^{\beta}{}_{\gamma},$	$\varepsilon_{\mathrm{RC}} := \xi^{\alpha} \Sigma_{\alpha} + (e_{\beta} \rfloor \widehat{D} \xi^{\gamma}) \tau^{\beta}{}_{\gamma},$	
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: basis] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_103 EQ0305 (now)	103	—	$\backslash \mathrm{varepsilon}_{\mathrm{RC}}:=\backslash \mathrm{x}\mathrm{i}^{\mathrm{\alpha}}\backslash \mathrm{Sigma}_{\mathrm{\alpha}}+\backslash \mathrm{left}(e_{\mathrm{\beta}}\backslash \mathrm{rfl oor}\overparen{\mathrm{D}}\backslash \mathrm{x}\mathrm{i}^{\mathrm{\gamma}}\mathrm{\right)}\backslash \mathrm{tau}^{\mathrm{\beta}}\mathrm{\beta}\mathrm{\gamma}\mathrm{\right)}\backslash \mathrm{tau}^{\mathrm{\beta}}\mathrm{\beta}\mathrm{\gamma}\mathrm{\right)}$	$\varepsilon_{\mathrm{RC}}:=\xi^{\alpha}\Sigma_{\alpha}+\left(e_{\beta}\rfloor\widehat{D}\xi^{\gamma}\right)\tau^{\beta}_{\gamma},$	$\varepsilon_{\mathrm{RC}}:=\xi^{\alpha}\Sigma_{\alpha}+(e_{\beta}\rfloor\widehat{D}\xi^{\gamma})\tau^{\beta}_{\gamma},$
basis: census / census					
mielke-geometrodynamics_103 EQ0306 (was)	103	■ 0.406	$\backslash \mathrm{left}.\mathrm{\widetilde{\varepsilon}}_{\mathrm{MA}}:=\backslash \mathrm{varepsilon}_{\mathrm{RC}}+\backslash \mathrm{x}\mathrm{i}\mathrm{\right)}\mathrm{\lrcorner}L\backslash \mathrm{cong}-\backslash \mathrm{left}(\mathrm{\mathscr{L}}_{\mathrm{\xi}}\backslash \mathrm{Gamma}_{\mathrm{\beta}}\mathrm{\right)}\mathrm{\right)}\mathrm{\wedge}H^{\mathrm{\beta}}\mathrm{\beta}\mathrm{\gamma}\mathrm{\right)}\mathrm{\wedge}H^{\mathrm{\beta}}\mathrm{\beta}\mathrm{\gamma}\mathrm{\right)}\mathrm{\wedge}H^{\mathrm{\beta}}\mathrm{\beta}\mathrm{\gamma}\mathrm{\right)}\mathrm{\wedge}H^{\mathrm{\beta}}\mathrm{\beta}\mathrm{\gamma}\mathrm{\right)}$	$\widetilde{\varepsilon}_{\mathrm{MA}}:=\varepsilon_{\mathrm{RC}}+\xi\lrcorner L\cong-(\mathscr{L}_{\xi}\Gamma_{\beta}^{\gamma})\wedge H^{\beta}_{\gamma}+dH_{\infty}$	$\widetilde{\varepsilon}_{\mathrm{MA}}:=\varepsilon_{\mathrm{RC}}+\xi\rfloor L\cong-(\mathscr{L}_{\xi}\Gamma_{\beta}^{\gamma})\wedge H^{\beta}_{\gamma}+dH_{\infty}$
mielke-geometrodynamics_103 EQ0306 (now)	103	—	$\mathrm{\widetilde{\varepsilon}}_{\mathrm{MA}}:=\mathrm{varepsilon}_{\mathrm{RC}}+\mathrm{x}\mathrm{i}\mathrm{\right)}\mathrm{\lrcorner}L\backslash \mathrm{cong}-\backslash \mathrm{left}(\mathrm{\mathscr{L}}_{\mathrm{\xi}}\backslash \mathrm{Gamma}_{\mathrm{\beta}}\mathrm{\right)}\mathrm{\right)}\mathrm{\wedge}H^{\mathrm{\beta}}\mathrm{\beta}\mathrm{\gamma}\mathrm{\right)}\mathrm{\wedge}H^{\mathrm{\beta}}\mathrm{\beta}\mathrm{\gamma}\mathrm{\right)}\mathrm{\wedge}H^{\mathrm{\beta}}\mathrm{\beta}\mathrm{\gamma}\mathrm{\right)}\mathrm{\wedge}H^{\mathrm{\beta}}\mathrm{\beta}\mathrm{\gamma}\mathrm{\right)}$	$\widetilde{\varepsilon}_{\mathrm{MA}}:=\varepsilon_{\mathrm{RC}}+\xi\rfloor L\cong-(\mathscr{L}_{\xi}\Gamma_{\beta}^{\gamma})\wedge H^{\beta}_{\gamma}+dH_{\infty}$	$\widetilde{\varepsilon}_{\mathrm{MA}}:=\varepsilon_{\mathrm{RC}}+\xi\rfloor L\cong-(\mathscr{L}_{\xi}\Gamma_{\beta}^{\gamma})\wedge H^{\beta}_{\gamma}+dH_{\infty}$
basis: census / census+ink					
mielke-geometrodynamics_104 EQ0311 (was)	104	■ 0.111	$\begin{aligned}&\mathrm{\widetilde{\leftarrow}}_{\mathrm{\widetilde{y}}}\mathrm{\widetilde{D}}\mathrm{\widetilde{\xi}}^{\alpha}=\mathrm{\widetilde{y}}\rfloor\left(\mathrm{\widetilde{D}}\mathrm{\widetilde{D}}\mathrm{\widetilde{\xi}}^{\alpha}\right)+\mathrm{\widetilde{D}}\left(\mathrm{\widetilde{y}}\rfloor\mathrm{\widetilde{D}}\mathrm{\widetilde{\xi}}^{\alpha}\right)\\&=yJ\left(DD\xi^{\alpha}+R^{(T)\alpha}\right)=0,\end{aligned}$	$\widetilde{\leftarrow}_{\widetilde{y}}\widetilde{D}\widetilde{\xi}^{\alpha}=\widetilde{y}\rfloor\left(\widetilde{D}\widetilde{D}\widetilde{\xi}^{\alpha}\right)+\widetilde{D}\left(\widetilde{y}\rfloor\widetilde{D}\widetilde{\xi}^{\alpha}\right)=y\rfloor(DD\xi^{\alpha}+R^{(T)\alpha})=0,$	$\widetilde{\mathbf{L}}_{\widetilde{y}}^{\widetilde{\approx}}\widetilde{D}\widetilde{\xi}^{\alpha}=\widetilde{y}\rfloor\left(\widetilde{D}\widetilde{D}\widetilde{\xi}^{\alpha}\right)+\widetilde{D}\left(\widetilde{y}\rfloor\widetilde{D}\widetilde{\xi}^{\alpha}\right)=y\rfloor(DD\xi^{\alpha}+R^{(T)\alpha})=0,$
mielke-geometrodynamics_104 EQ0311 (now)	104	—	$\begin{aligned}&\mathrm{\widetilde{\leftarrow}}_{\mathrm{\widetilde{y}}}\mathrm{\widetilde{D}}\mathrm{\widetilde{\xi}}^{\alpha}=\mathrm{\widetilde{y}}\rfloor\left(\mathrm{\widetilde{D}}\mathrm{\widetilde{D}}\mathrm{\widetilde{\xi}}^{\alpha}\right)+\mathrm{\widetilde{D}}\left(\mathrm{\widetilde{y}}\rfloor\mathrm{\widetilde{D}}\mathrm{\widetilde{\xi}}^{\alpha}\right)\\&=yJ\left(DD\xi^{\alpha}+R^{(T)\alpha}\right)=0,\end{aligned}$	$\widetilde{\leftarrow}_{\widetilde{y}}\widetilde{D}\widetilde{\xi}^{\alpha}=\widetilde{y}\rfloor\left(\widetilde{D}\widetilde{D}\widetilde{\xi}^{\alpha}\right)+\widetilde{D}\left(\widetilde{y}\rfloor\widetilde{D}\widetilde{\xi}^{\alpha}\right)=y\rfloor(DD\xi^{\alpha}+R^{(T)\alpha})=0,$	$\widetilde{\mathbf{L}}_{\widetilde{y}}^{\widetilde{\approx}}\widetilde{D}\widetilde{\xi}^{\alpha}=\widetilde{y}\rfloor\left(\widetilde{D}\widetilde{D}\widetilde{\xi}^{\alpha}\right)+\widetilde{D}\left(\widetilde{y}\rfloor\widetilde{D}\widetilde{\xi}^{\alpha}\right)=y\rfloor(DD\xi^{\alpha}+R^{(T)\alpha})=0,$
basis: census / census					
mielke-geometrodynamics_104 EQ0312 (was)	104	■ 0.786	$\begin{aligned}&\Delta\xi^{\alpha}=-\oint_Cy\mathrm{\right)}\left(DD\xi^{\alpha}\right)=\oint_Cy\mathrm{\right)}R^{(T)\alpha}=\int_SR^{(T)\alpha}\\&\simeq\frac{1}{2}\left(T_{ij}^{\alpha}-R_{ij\beta}^{\alpha}\xi^{\beta}\right)\int_Sdy^i\wedge dy^j\end{aligned}$	$\Delta\xi^{\alpha}=-\oint_Cy\rfloor(DD\xi^{\alpha})=\oint_Cy\rfloor R^{(T)\alpha}=\int_S R^{(T)\alpha}\simeq\frac{1}{2}\left(T_{ij}^{\alpha}-R_{ij\beta}^{\alpha}\xi^{\beta}\right)\int_S dy^i\wedge dy^j.$	$\Delta\xi^{\alpha}=-\oint_Cy\rfloor(DD\xi^{\alpha})=\oint_Cy\rfloor R^{(T)\alpha}=\int_S R^{(T)\alpha}\simeq\frac{1}{2}\left(T_{ij}^{\alpha}-R_{ij\beta}^{\alpha}\xi^{\beta}\right)\int_S dy^i\wedge dy^j.$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5-0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_104 E00312 (now)	104	—	$\begin{aligned} \Delta \xi^\alpha &= - \oint_C y \rfloor (DD\xi^\alpha) = \oint_C y \rfloor R^{(T)\alpha} = \int_S R^{(T)\alpha} \\ &\simeq \frac{1}{2} (T_{ij}^\alpha - R_{ij\beta}^\alpha \xi^\beta) \int_S dy^i \wedge dy^j \end{aligned}$	$\Delta \xi^\alpha = - \oint_C y \rfloor (DD\xi^\alpha) = \oint_C y \rfloor R^{(T)\alpha} = \int_S R^{(T)\alpha}$	$\Delta \xi^\alpha = - \oint_C y \rfloor (DD\xi^\alpha) = \oint_C y \rfloor R^{(T)\alpha} = \int_S R^{(T)\alpha}$$\simeq \frac{1}{2} (T_{ij}^\alpha - R_{ij\beta}^\alpha \xi^\beta) \int_S dy^i \wedge dy^j .$
basis: census / census+ink					
mielke-geometrodynamics_111 E00322 (was)	111	■ 1.000	$\left. dU = T^\alpha \wedge \mu_\alpha - \vartheta^\alpha \wedge D\mu_\alpha = \vartheta^\alpha \wedge [e_\beta] (T^\beta \wedge \mu_\alpha) - D\mu_\alpha \right _{\left. \left[e_\beta \wedge \mu_\alpha = \vartheta^\alpha \wedge D\mu_\alpha \right] \right _{\left[e_\beta \wedge \mu_\alpha = \vartheta^\alpha \wedge D\mu_\alpha \right]}}$	$dU = T^\alpha \wedge \mu_\alpha - \vartheta^\alpha \wedge D\mu_\alpha = \vartheta^\alpha \wedge [e_\beta] (T^\beta \wedge \mu_\alpha) - D\mu_\alpha$	$dU = T^\alpha \wedge \mu_\alpha - \vartheta^\alpha \wedge D\mu_\alpha = \vartheta^\alpha \wedge [e_\beta] (T^\beta \wedge \mu_\alpha) - D\mu_\alpha$
mielke-geometrodynamics_111 E00322 (now)	111	—	$dU = T^\alpha \wedge \mu_\alpha - \vartheta^\alpha \wedge D\mu_\alpha = \vartheta^\alpha \wedge [e_\beta] (T^\beta \wedge \mu_\alpha) - D\mu_\alpha$	$dU = T^\alpha \wedge \mu_\alpha - \vartheta^\alpha \wedge D\mu_\alpha = \vartheta^\alpha \wedge [e_\beta] (T^\beta \wedge \mu_\alpha) - D\mu_\alpha$	$dU = T^\alpha \wedge \mu_\alpha - \vartheta^\alpha \wedge D\mu_\alpha = \vartheta^\alpha \wedge [e_\beta] (T^\beta \wedge \mu_\alpha) - D\mu_\alpha$
basis: census / census+ink					
mielke-geometrodynamics_111 E00323 (was)	111	■ 1.000	$\begin{aligned} \Sigma_\alpha &\rightarrow \sigma_\alpha := \Sigma_\alpha - D\mu_\alpha + e_\beta \rfloor (T^\beta \wedge \mu_\alpha) , \\ \tau_{\alpha\beta} &\rightarrow \hat{\tau}_{\alpha\beta} := \tau_{\alpha\beta} - \vartheta_{[\alpha} \wedge \mu_{\beta]} = 0 \end{aligned}$	$\Sigma_\alpha \rightarrow \sigma_\alpha := \Sigma_\alpha - D\mu_\alpha + e_\beta \rfloor (T^\beta \wedge \mu_\alpha) ,$$\tau_{\alpha\beta} \rightarrow \hat{\tau}_{\alpha\beta} := \tau_{\alpha\beta} - \vartheta_{[\alpha} \wedge \mu_{\beta]} = 0$	$\Sigma_\alpha \rightarrow \sigma_\alpha := \Sigma_\alpha - D\mu_\alpha + e_\beta \rfloor (T^\beta \wedge \mu_\alpha) ,$$\tau_{\alpha\beta} \rightarrow \hat{\tau}_{\alpha\beta} := \tau_{\alpha\beta} - \vartheta_{[\alpha} \wedge \mu_{\beta]} = 0$
mielke-geometrodynamics_111 E00323 (now)	111	—	$\begin{aligned} \Sigma_\alpha &\rightarrow \sigma_\alpha := \Sigma_\alpha - D\mu_\alpha + e_\beta \rfloor (T^\beta \wedge \mu_\alpha) , \\ \tau_{\alpha\beta} &\rightarrow \hat{\tau}_{\alpha\beta} := \tau_{\alpha\beta} - \vartheta_{[\alpha} \wedge \mu_{\beta]} = 0 \end{aligned}$	$\Sigma_\alpha \rightarrow \sigma_\alpha := \Sigma_\alpha - D\mu_\alpha + e_\beta \rfloor (T^\beta \wedge \mu_\alpha) ,$$\tau_{\alpha\beta} \rightarrow \hat{\tau}_{\alpha\beta} := \tau_{\alpha\beta} - \vartheta_{[\alpha} \wedge \mu_{\beta]} = 0$	$\Sigma_\alpha \rightarrow \sigma_\alpha := \Sigma_\alpha - D\mu_\alpha + e_\beta \rfloor (T^\beta \wedge \mu_\alpha) ,$$\tau_{\alpha\beta} \rightarrow \hat{\tau}_{\alpha\beta} := \tau_{\alpha\beta} - \vartheta_{[\alpha} \wedge \mu_{\beta]} = 0$
basis: census / census+ink					
mielke-geometrodynamics_112 E00326 (was)	112	■ 0.613	$\left. \widehat{D} G_\alpha \equiv \frac{1}{2} (e_\alpha \rfloor R^{\beta\gamma}) \wedge \eta_{\beta\gamma\mu} \wedge T^\mu \right _{\left. \left(e_\alpha \rfloor R^{\beta\gamma} \right) \wedge \eta_{\beta\gamma\mu} \wedge T^\mu \right _{\left(e_\alpha \rfloor R^{\beta\gamma} \right) \wedge \eta_{\beta\gamma\mu} \wedge T^\mu}}$	$\widehat{D} G_\alpha \equiv \frac{1}{2} (e_\alpha \rfloor R^{\beta\gamma}) \wedge \eta_{\beta\gamma\mu} \wedge T^\mu$	$\widehat{D} G_\alpha \equiv \frac{1}{2} (e_\alpha \rfloor R^{\beta\gamma}) \wedge \eta_{\beta\gamma\mu} \wedge T^\mu$
mielke-geometrodynamics_112 E00326 (now)	112	—	$\widehat{D} G_\alpha \equiv \frac{1}{2} (e_\alpha \rfloor R^{\beta\gamma}) \wedge \eta_{\beta\gamma\mu} \wedge T^\mu$	$\widehat{D} G_\alpha \equiv \frac{1}{2} (e_\alpha \rfloor R^{\beta\gamma}) \wedge \eta_{\beta\gamma\mu} \wedge T^\mu$	$\widehat{D} G_\alpha \equiv \frac{1}{2} (e_\alpha \rfloor R^{\beta\gamma}) \wedge \eta_{\beta\gamma\mu} \wedge T^\mu$
basis: census / census+ink					
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value					
LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ					
Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured					
[refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_115 E00340 (was)		■ 0.962	$\begin{aligned} G_{\alpha} & \& := \frac{1}{2} R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \\ & = G_{\alpha}^{\{\}} + \frac{(-1)^s}{12} \left(e_{\alpha} \rfloor \mathcal{A} \wedge {}^* \mathcal{A} - \frac{1}{3} \mathcal{A} \wedge e_{\alpha} \rfloor {}^* \mathcal{A} \right) + \frac{(-1)^s}{6} \vartheta_{\alpha} \wedge d\mathcal{A} \end{aligned}$	$G_{\alpha} := \frac{1}{2} R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma}$ $= G_{\alpha}^{\{\}} + \frac{(-1)^s}{12} \left(e_{\alpha} \rfloor \mathcal{A} \wedge {}^* \mathcal{A} - \frac{1}{3} \mathcal{A} \wedge e_{\alpha} \rfloor {}^* \mathcal{A} \right) + \frac{(-1)^s}{6} \vartheta_{\alpha} \wedge d\mathcal{A}$	$G_{\alpha} := \frac{1}{2} R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \tag{5.3}$ $= G_{\alpha}^{\{\}} + \frac{(-1)^s}{12} \left(e_{\alpha} \rfloor \mathcal{A} \wedge {}^* \mathcal{A} - \frac{1}{3} \mathcal{A} \wedge e_{\alpha} \rfloor {}^* \mathcal{A} \right) + \frac{(-1)^s}{6} \vartheta_{\alpha} \wedge d\mathcal{A}$
mielke-geometrodynamics_115 E00340 (now)		—	$\begin{aligned} G_{\alpha} & \& := \frac{1}{2} R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \\ & = G_{\alpha}^{\{\}} + \frac{(-1)^s}{12} \left(e_{\alpha} \rfloor \mathcal{A} \wedge {}^* \mathcal{A} - \frac{1}{3} \mathcal{A} \wedge e_{\alpha} \rfloor {}^* \mathcal{A} \right) + \frac{(-1)^s}{6} \vartheta_{\alpha} \wedge d\mathcal{A} \end{aligned}$	$G_{\alpha} := \frac{1}{2} R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma}$ $= G_{\alpha}^{\{\}} + \frac{(-1)^s}{12} \left(e_{\alpha} \rfloor \mathcal{A} \wedge {}^* \mathcal{A} - \frac{1}{3} \mathcal{A} \wedge e_{\alpha} \rfloor {}^* \mathcal{A} \right) + \frac{(-1)^s}{6} \vartheta_{\alpha} \wedge d\mathcal{A}$	$G_{\alpha} := \frac{1}{2} R^{\beta\gamma} \wedge \eta_{\alpha\beta\gamma} \tag{5.3}$ $= G_{\alpha}^{\{\}} + \frac{(-1)^s}{12} \left(e_{\alpha} \rfloor \mathcal{A} \wedge {}^* \mathcal{A} - \frac{1}{3} \mathcal{A} \wedge e_{\alpha} \rfloor {}^* \mathcal{A} \right) + \frac{(-1)^s}{6} \vartheta_{\alpha} \wedge d\mathcal{A}$
basis: census / census+ink					
mielke-geometrodynamics_127 E00372 (was)		■ 0.437	$\left. \left(\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} \mp \frac{i}{4} \ell^2 e_{\alpha} \right) \right(\stackrel{(+)}{\Pi}^{\beta} \wedge \stackrel{(-)}{\Pi}_{\beta}) = \Sigma_{\alpha},$	$\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} \mp \frac{i}{4} \ell^2 e_{\alpha} \rfloor \left(\stackrel{(+)}{\Pi}^{\beta} \wedge \stackrel{(-)}{\Pi}_{\beta} \right) = \Sigma_{\alpha},$	$\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} \mp \frac{i}{4} \ell^2 e_{\alpha} \rfloor \left(\stackrel{(+)}{\Pi}^{\beta} \wedge \stackrel{(-)}{\Pi}_{\beta} \right) = \Sigma_{\alpha},$
mielke-geometrodynamics_127 E00372 (now)		—	$\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} \mp \frac{i}{4} \ell^2 e_{\alpha} \rfloor \left(\stackrel{(+)}{\Pi}^{\beta} \wedge \stackrel{(-)}{\Pi}_{\beta} \right) = \Sigma_{\alpha},$	$\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} \mp \frac{i}{4} \ell^2 e_{\alpha} \rfloor \left(\stackrel{(+)}{\Pi}^{\beta} \wedge \stackrel{(-)}{\Pi}_{\beta} \right) = \Sigma_{\alpha},$	$\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} \mp \frac{i}{4} \ell^2 e_{\alpha} \rfloor \left(\stackrel{(+)}{\Pi}^{\beta} \wedge \stackrel{(-)}{\Pi}_{\beta} \right) = \Sigma_{\alpha},$
basis: census / census+ink					
mielke-geometrodynamics_128 E00378 (was)		■ 0.136	$D \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} \mp \frac{i}{4} \ell^2 e_{\alpha} \rfloor \left(\stackrel{(+)}{\Pi}^{\beta} \wedge \stackrel{(-)}{\Pi}_{\beta} \right) = \Sigma_{\alpha},$	$DD\vartheta^{\alpha} = R_{\beta}^{\alpha} \wedge \vartheta^{\beta} = 0, \stackrel{(\pm)}{D} \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} = - \stackrel{(\pm)}{R}_{\alpha}^{\beta} \wedge \stackrel{(\pm)}{\Pi}_{\beta} = \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \rfloor \stackrel{(\pm)}{\Pi}^{\beta} \right) \wedge \Sigma_{\beta} \cong 0.$	$DD\vartheta^{\alpha} = R_{\beta}^{\alpha} \wedge \vartheta^{\beta} = 0, \stackrel{(\pm)}{D} \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} = - \stackrel{(\pm)}{R}_{\alpha}^{\beta} \wedge \stackrel{(\pm)}{\Pi}_{\beta} = \pm \frac{i}{2} \ell^2 (e_{\alpha} \rfloor \stackrel{(\pm)}{\Pi}^{\beta}) \wedge \Sigma_{\beta} \cong 0.$
mielke-geometrodynamics_128 E00378 (now)		—	$DD\vartheta^{\alpha} = R_{\beta}^{\alpha} \wedge \vartheta^{\beta} = 0, \stackrel{(\pm)}{D} \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} = - \stackrel{(\pm)}{R}_{\alpha}^{\beta} \wedge \stackrel{(\pm)}{\Pi}_{\beta} = \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \rfloor \stackrel{(\pm)}{\Pi}^{\beta} \right) \wedge \Sigma_{\beta} \cong 0.$	$DD\vartheta^{\alpha} = R_{\beta}^{\alpha} \wedge \vartheta^{\beta} = 0, \stackrel{(\pm)}{D} \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} = - \stackrel{(\pm)}{R}_{\alpha}^{\beta} \wedge \stackrel{(\pm)}{\Pi}_{\beta} = \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \rfloor \stackrel{(\pm)}{\Pi}^{\beta} \right) \wedge \Sigma_{\beta} \cong 0.$	$DD\vartheta^{\alpha} = R_{\beta}^{\alpha} \wedge \vartheta^{\beta} = 0, \stackrel{(\pm)}{D} \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} = - \stackrel{(\pm)}{R}_{\alpha}^{\beta} \wedge \stackrel{(\pm)}{\Pi}_{\beta} = \pm \frac{i}{2} \ell^2 (e_{\alpha} \rfloor \stackrel{(\pm)}{\Pi}^{\beta}) \wedge \Sigma_{\beta} \cong 0.$
basis: census / census					
mielke-geometrodynamics_129 E00381 (was)		■ 0.588	$\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} := \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \rfloor \stackrel{(\mp)}{\Pi}^{\beta} \right) \wedge \stackrel{(\pm)}{\Pi}_{\beta},$	$\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} := \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} \pm \frac{i}{2} \ell^2 \left(e_{\alpha} \rfloor \stackrel{(\mp)}{\Pi}^{\beta} \right) \wedge \stackrel{(\pm)}{\Pi}_{\beta},$	$\stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} := \stackrel{(\pm)}{D} \stackrel{(\pm)}{\Pi}_{\alpha} \pm \frac{i}{2} \ell^2 (e_{\alpha} \rfloor \stackrel{(\mp)}{\Pi}^{\beta}) \wedge \stackrel{(\pm)}{\Pi}_{\beta},$
Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ● C component ● W weak ● S stable ● N noise ● K clean ● not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

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mielke-geometrodynamics_129 EQ0381 (now)	129	—	$\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_\alpha:=D\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_\alpha\pm\frac{i}{2}\ell^2\left(e_\alpha\rfloor\stackrel{\scriptscriptstyle{(\mp)}}{\Pi}\right)\wedge\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_\beta,$	$\stackrel{\scriptscriptstyle{(\pm)}}{D}\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_\alpha:=D\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_\alpha\pm\frac{i}{2}\ell^2\left(e_\alpha\rfloor\stackrel{\scriptscriptstyle{(\mp)}}{\Pi}\right)\wedge\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_\beta,$	$\stackrel{\scriptscriptstyle{(\pm)}}{D}\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_\alpha:=D\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_\alpha\pm\frac{i}{2}\ell^2\left(e_\alpha\rfloor\stackrel{\scriptscriptstyle{(\mp)}}{\Pi}^\beta\right)\wedge\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}_\beta,$
basis: census / census					
mielke-geometrodynamics_129 EQ0382 (was)	129	■0.087	$\stackrel{\scriptscriptstyle{(\pm)}}{\Gamma}_\alpha{}^\beta:=\Gamma_\alpha{}^\beta+i\stackrel{\scriptscriptstyle{(\mp)}}{A}^\beta{}_\beta,\quad\stackrel{\scriptscriptstyle{(\pm)}}{A}^\beta{}_\beta:=\pm\frac{1}{2}\ell^2e_\alpha\rfloor\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}^\beta,$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Gamma}_\alpha{}^\beta:=\Gamma_\alpha{}^\beta+i\stackrel{\scriptscriptstyle{(\mp)}}{A}^\beta{}_\beta,\quad\stackrel{\scriptscriptstyle{(\pm)}}{A}^\beta{}_\beta:=\pm\frac{1}{2}\ell^2e_\alpha\rfloor\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}^\beta,$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Gamma}_\alpha{}^\beta:=\Gamma_\alpha{}^\beta+i\stackrel{\scriptscriptstyle{(\mp)}}{A}^\beta{}_\beta,\quad\stackrel{\scriptscriptstyle{(\pm)}}{A}^\beta{}_\beta:=\pm\frac{1}{2}\ell^2e_\alpha\rfloor\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}^\beta,$
mielke-geometrodynamics_129 EQ0382 (now)	129	—	$\stackrel{\scriptscriptstyle{(\pm)}}{\Gamma}_\alpha{}^\beta:=\Gamma_\alpha{}^\beta+i\stackrel{\scriptscriptstyle{(\mp)}}{A}^\beta{}_\beta,\quad\stackrel{\scriptscriptstyle{(\pm)}}{A}^\beta{}_\beta:=\pm\frac{1}{2}\ell^2e_\alpha\rfloor\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}^\beta,$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Gamma}_\alpha{}^\beta:=\Gamma_\alpha{}^\beta+i\stackrel{\scriptscriptstyle{(\mp)}}{A}^\beta{}_\beta,\quad\stackrel{\scriptscriptstyle{(\pm)}}{A}^\beta{}_\beta:=\pm\frac{1}{2}\ell^2e_\alpha\rfloor\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}^\beta,$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Gamma}_\alpha{}^\beta:=\Gamma_\alpha{}^\beta+i\stackrel{\scriptscriptstyle{(\mp)}}{A}^\beta{}_\beta,\quad\stackrel{\scriptscriptstyle{(\pm)}}{A}^\beta{}_\beta:=\pm\frac{1}{2}\ell^2e_\alpha\rfloor\stackrel{\scriptscriptstyle{(\pm)}}{\Pi}^\beta,$
basis: census / census					
mielke-geometrodynamics_130 EQ0387 (was)	130	■0.994	$\left.\stackrel{\scriptscriptstyle{(\pm)}}{\Psi}\right\rangle:=dt\wedge\Psi_\perp;\quad\Psi_\perp:=n\rfloor\Psi,$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Psi}:=dt\wedge\Psi_\perp;\quad\Psi_\perp:=n\rfloor\Psi,$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Psi}:=dt\wedge\Psi_\perp;\quad\Psi_\perp:=n\rfloor\Psi,$
mielke-geometrodynamics_130 EQ0387 (now)	130	—	$\left.\stackrel{\scriptscriptstyle{(\pm)}}{\Psi}\right\rangle:=dt\wedge\Psi_\perp;\quad\Psi_\perp:=n\rfloor\Psi,$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Psi}:=dt\wedge\Psi_\perp;\quad\Psi_\perp:=n\rfloor\Psi,$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Psi}:=dt\wedge\Psi_\perp;\quad\Psi_\perp:=n\rfloor\Psi,$
basis: census / census+ink					
mielke-geometrodynamics_130 EQ0388 (was)	130	■0.993	$\left.\stackrel{\scriptscriptstyle{(\pm)}}{\Psi}\right\rangle:=dt\wedge\Psi_\perp;\quad\Psi_\perp:=n\rfloor\Psi,$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Psi}:=dt\wedge\Psi_\perp;\quad\Psi_\perp:=n\rfloor\Psi,$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Psi}:=dt\wedge\Psi_\perp;\quad\Psi_\perp:=n\rfloor\Psi,$
mielke-geometrodynamics_130 EQ0388 (now)	130	—	$\left.\stackrel{\scriptscriptstyle{(\pm)}}{\Psi}\right\rangle:=dt\wedge\Psi_\perp;\quad\Psi_\perp:=n\rfloor\Psi,$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Psi}:=dt\wedge\Psi_\perp;\quad\Psi_\perp:=n\rfloor\Psi,$	$\stackrel{\scriptscriptstyle{(\pm)}}{\Psi}:=dt\wedge\Psi_\perp;\quad\Psi_\perp:=n\rfloor\Psi,$
basis: census / census+ink					
mielke-geometrodynamics_131 EQ0391 (was)	131	■0.066	$\begin{aligned}\Sigma_\alpha&:=\frac{\delta L}{\delta\vartheta^\alpha}=e_\alpha\rfloor L-(e_\alpha\rfloor D\Psi)\wedge\frac{\partial L}{\partial D\Psi}-(e_\alpha\rfloor\Psi)\wedge\frac{\partial L}{\partial\Psi}\\&\quad-(e_\alpha\rfloor T^\beta)\wedge\frac{\partial L}{\partial T^\beta}+D\frac{\partial L}{\partial T^\alpha}-(e_\alpha\rfloor R_\beta{}^\gamma)\wedge\frac{\partial L}{\partial R_\beta{}^\gamma}.\end{aligned}$	$\Sigma_\alpha:=\frac{\delta L}{\delta\vartheta^\alpha}=e_\alpha\rfloor L-(e_\alpha\rfloor D\Psi)\wedge\frac{\partial L}{\partial D\Psi}-(e_\alpha\rfloor\Psi)\wedge\frac{\partial L}{\partial\Psi}\\-(e_\alpha\rfloor T^\beta)\wedge\frac{\partial L}{\partial T^\beta}+D\frac{\partial L}{\partial T^\alpha}-(e_\alpha\rfloor R_\beta{}^\gamma)\wedge\frac{\partial L}{\partial R_\beta{}^\gamma}.$	$\Sigma_\alpha:=\frac{\delta L}{\delta\vartheta^\alpha}=e_\alpha\rfloor L-(e_\alpha\rfloor D\Psi)\wedge\frac{\partial L}{\partial D\Psi}-(e_\alpha\rfloor\Psi)\wedge\frac{\partial L}{\partial\Psi}\\-(e_\alpha\rfloor T^\beta)\wedge\frac{\partial L}{\partial T^\beta}+D\frac{\partial L}{\partial T^\alpha}-(e_\alpha\rfloor R_\beta{}^\gamma)\wedge\frac{\partial L}{\partial R_\beta{}^\gamma}.$
Conf. MathPix confidence: ■≥0.9 ■0.5–0.9 ■<0.5 — no value LaTeX source MathPix’s reading, unchanged; Rendered the same reading, with a display environment (align/alignat/flalign/eqnarray/gather/multline/split) mapped to its in-math form so it can typeset in this cell — the two may legitimately differ Residual render vs scan (inkdrill): ●C component ●W weak ●S stable ●N noise ●K clean ●not measured [refined: <i>basis</i>] beside an identifier: the LaTeX-source and Rendered cells both show a VERIFIED refinement, not MathPix’s own reading — <i>basis</i> names what verified it (e.g. ink, source). The original reading is unchanged in the model but is not the one drawn here.					

